

I. Evaluation

We evaluate EsCapturer-full through six research questions. The primary quantitative evidence comes from two independent 50,000-record balanced benchmarks: one built from Quo Vadis and one built from Zenodo 11079764. A stricter zero-overlap audit tests split reliability, followed by matched ablation, robustness, efficiency, and gate-weight analyses.

A. Experimental Setup

1) Research Questions: RQ1: Overall Detection Performance. Does EsCapturer-full accurately distinguish malware from benign API/syscall traces? RQ2: Split Reliability. How does EsCapturer-full behave after exact deduplication and source-group-disjoint partitioning? RQ3: Ablation Study. What are the marginal contributions of semantic elements, the graph view, and the interleaving prior? RQ4: Robustness. How stable is EsCapturer-full under API/syscall insertion, deletion, and local reordering? RQ5: Efficiency and Deployment Overhead. What are the costs of extraction, behavior-unit construction, graph construction, training, and inference? RQ6: Interpretability. Do the learned view weights agree with the interleaving prior, and can they explain when the model prefers sequence or graph representations?

2) Datasets and Splits: Quo Vadis contains report-level ordered Windows API sequences produced by Speakeasy; each report is capped at 5,000 calls. Zenodo record 11079764 contains VMI API traces; each trace is divided into windows of at most 200 calls with stride 100. For the main benchmark, each dataset contributes 50,000 records, balanced between 25,000 benign and 25,000 malware records, with its own seed-7 35,000/7,500/7,500 train/validation/test split. Thus, the Quo Vadis result and the Zenodo result are two separate 50,000-record evaluations, not a single mixed 50,000-record corpus. We report Accuracy, Precision, Recall, F1, and ROC-AUC for the malicious class, using validation data for checkpoint and threshold selection.

As a reliability audit, we additionally construct stricter partitions by removing exact API sequences that appear with conflicting labels, collapsing remaining exact duplicates, assigning report/SHA groups atomically to train/validation/test splits, and auditing exact-sequence and source-group overlap across all partition pairs. This procedure yields 6,636 Quo Vadis records and 38,882 Zenodo records, with zero exact-sequence overlap and zero report/SHA group overlap between train, validation, and test. We use this audit to check data-split reliability rather than as the main performance ranking table.

Dataset	Final records	Exact overlap	Group overlap
Quo Vadis	6,636	0	0
Zenodo 11079764	38,882	0	0

3) Baselines and Implementation: All reported baseline scores are local runs under the same benchmark split and metric definitions, rather than numbers copied from prior

papers. We compare against representative frequency, sequence, graph, and adapted malware-detection systems, including API-frequency and API n -gram classifiers, Nebula, API2Vec++, DawnGNN, GPT4-API-BERT-CNN, DeepCapa, APILI, and MME-TextCNN. EsCapturer-full uses a frozen semantic-template extractor, sequence and graph views, and interleaving-prior adaptive fusion. Vocabularies, caches, normalizers, and graph statistics are fitted on training records only. The semantic-template extractor is frozen and cached, so training and deployment-time inference do not make online LLM calls.

4) Training and Selection: The balanced Full model and matched ablations use a BCE objective, 32-dimensional embeddings, learning rate 5×10^{-4} , at most 30 epochs, patience 5, maximum sequence length 512, and at most 16 behavior units selected by uniform coverage. Epoch-resampled balanced pairs expose all 17,500 benign training records per epoch and sample an equal number of attacks. Checkpoints maximize validation F1, and a separate validation sweep fixes the decision threshold before test inference. The selected Full thresholds are 0.6567 for Quo Vadis and 0.0700 for Zenodo; their different scales do not affect AUC and are never retuned on test data. RQ1, RQ3, RQ4, and RQ6 use seed 7, whereas RQ2 repeats training with seeds 7, 17, and 27 on one fixed zero-overlap partition.

5) Artifact Checks: Every balanced Full and matched-ablation row is backed by an independent checkpoint, configuration, metrics file, and 7,500 test predictions. Each RQ4 condition additionally stores the perturbed test records and 7,500 condition-specific predictions. The Quo Vadis zero-overlap runs contain 995 test predictions per seed. We recompute reported deltas from the corresponding clean baseline and exclude legacy-objective or incomplete runs from the paper-facing comparisons.

B. RQ1: Overall Detection Performance

Table I summarizes the two 50,000-record balanced benchmarks. The Quo Vadis columns are computed on the Quo Vadis 50,000-record split, and the Zenodo columns are computed on the separate Zenodo 11079764 50,000-record split. We use a compact F1/AUC table to emphasize the metrics most directly tied to detection effectiveness and threshold-independent ranking. For EsCapturer-full, the complete five-metric results are: on Quo Vadis, Accuracy 0.9631, Precision 0.9743, Recall 0.9512, F1 0.9626, and AUC 0.9935; on Zenodo 11079764, Accuracy 0.9405, Precision 0.9345, Recall 0.9475, F1 0.9409, and AUC 0.9791.

On Quo Vadis, EsCapturer-full achieves F1/AUC of 0.9626/0.9935 and the highest precision among the listed methods, while DeepCapa-adapted is 0.0022/0.0018 higher on F1/AUC. On Zenodo, EsCapturer-full obtains F1/AUC of 0.9409/0.9791; the TF-IDF/SGD prototype is 0.0107/0.0083 higher, and DeepCapa-adapted is 0.0043/0.0072 higher. These are local, task-adapted comparisons rather than claims about the original systems.

TABLE I

Local detection results on the two 50,000-record balanced benchmarks. Adapted and reimplemented methods use the common binary task.

Method	Quo F1	Quo AUC	Zenodo F1	Zenodo AUC
EsCapturer-full, repaired	0.9626	0.9935	0.9409	0.9791
TF-IDF/SGD prototype	0.9618	0.9947	0.9516	0.9874
Nebula	0.9135	0.9708	0.7921	0.8994
API2Vec++	0.9281	0.9824	0.9069	0.9611
DawnGNN-reimpl	0.8800	0.9458	0.8030	0.9189
GPT4-API-BERT-CNN	0.9544	0.9926	0.9228	0.9737
DeepCapa-adapted	0.9648	0.9953	0.9452	0.9863
APILI-adapted	0.9388	0.9869	0.9368	0.9806
MME-TextCNN	0.9389	0.9890	0.9287	0.9789

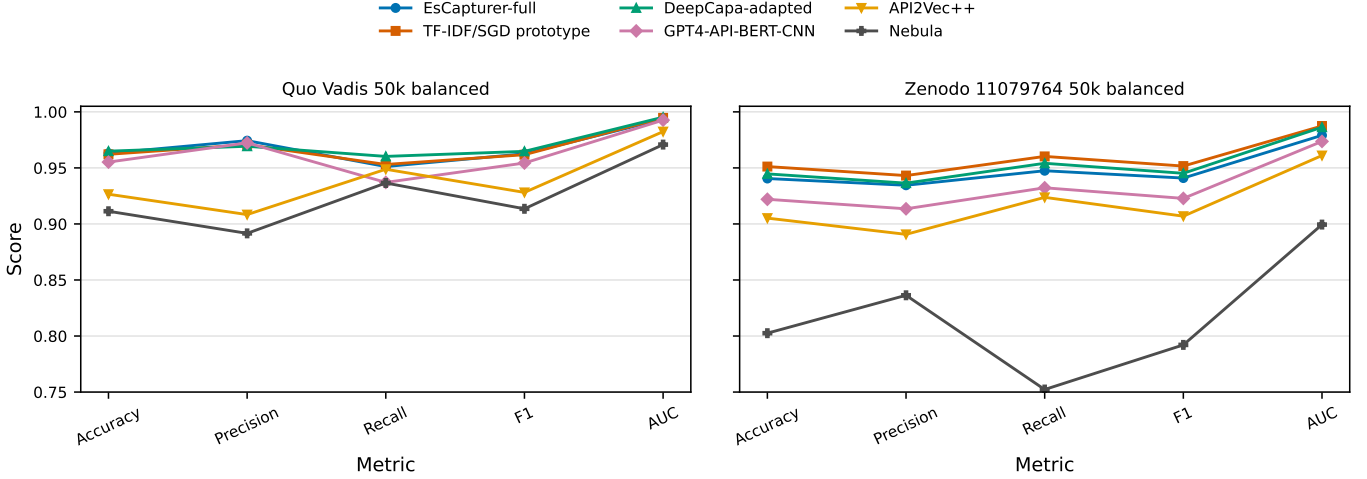


Fig. 1. Five-metric detection comparison on the Quo Vadis (left) and Zenodo 11079764 (right) 50,000-record balanced benchmarks. Each line represents one method with complete Accuracy, Precision, Recall, F1, and AUC results.

Answer to RQ1. EsCapturer-full provides strong and competitive malware/benign detection on both 50,000-record balanced benchmarks, while remaining close to the strongest local adapted baselines rather than uniformly exceeding them.

C. RQ2: Split Reliability

The balanced samples contain repeated sequences and records from related source reports or SHAs. Our stricter audit removes conflicting-label exact sequences, deduplicates the remainder, and assigns source groups atomically. The resulting Quo Vadis and Zenodo partitions contain 6,636 and 38,882 records, respectively, with zero exact-sequence and zero source-group overlap for every train/validation/test pair. Complete repaired-model results are available for three training seeds on the Quo Vadis audit split (4,647/994/995 records); checkpoints and thresholds are validation-selected for each seed.

The reconstruction removes 6,895 Quo Vadis rows involved in 129 conflicting-label sequence groups and then removes 36,469 duplicate rows, retaining 6,636 records. For Zenodo, it removes 3,601 rows in 145 conflicting groups and 7,517 additional duplicates, retaining 38,882 records. These counts explain why the audit is a different population rather than merely a new random split of the balanced benchmark.

TABLE II

Quo Vadis zero-overlap audit results. The final row reports mean and sample standard deviation across training seeds.

Seed	F1	AUC
7	0.7603	0.9029
17	0.7883	0.9225
27	0.7516	0.9249
Mean $\pm$ SD	0.7667 $\pm$ 0.0191	0.9168 $\pm$ 0.0121

The audit preserves a useful ranking signal but is substantially harder than the balanced split. Across seeds, Accuracy is 0.8911 ± 0.0065 , Precision 0.8453 ± 0.0047 , and Recall 0.7021 ± 0.0344 . Because the audit also changes class balance and removes repeated records, the difference from RQ1 should not be attributed to duplication alone.

Answer to RQ2. Exact- and group-disjoint partitioning reduces measured performance, but the three-seed Quo Vadis audit retains AUC 0.9168 ± 0.0121 . This confirms nontrivial discrimination under the stricter split while exposing the sensitivity of balanced-split estimates to sampling and source overlap.

D. RQ3: Ablation Study

RQ3 studies the contribution of EsCapturer-full’s semantic elements, graph view, and interleaving prior. Each variant uses the same 50,000-record balanced seed-7 split,

training budget, validation-based checkpoint selection, and validation-selected decision threshold as the full model. The compact comparison below reports F1 and its change from the corresponding repaired BCE Full model.

Variant	Quo F1	Δ_Q	Zenodo F1	Δ_Z
Full	0.9626	0	0.9409	0
w/o semantic elements	0.9633	+0.0007	0.9289	-0.0120
w/o graph view	0.9600	-0.0026	0.9281	-0.0128
w/o interleaving prior	0.9596	-0.0030	0.9422	+0.0013

All runs use the corresponding 35,000/7,500/7,500 seed-7 split, the full model’s training budget, validation-selected checkpoint, and validation-selected threshold. On Quo Vadis, replacing semantic behavior elements with matched raw API groups gives F1 0.9633, slightly above the full model’s 0.9626, so semantic elements provide no additional F1 gain on this balanced split. Removing the graph view and interleaving prior reduces F1 by 0.0026 and 0.0030, respectively, indicating small positive contributions from object/relation structure and the fusion prior in this setting. On Zenodo, removing semantic elements and the graph view reduces F1 by 0.0120 and 0.0128, respectively, showing clearer contributions from behavioral semantics and object/relation structure on this split. Removing the interleaving prior instead increases F1 by 0.0013, from 0.9409 to 0.9422. This result does not support a positive marginal F1 contribution from the prior on the Zenodo balanced split and motivates further examination through strict-split and interpretability checks.

The AUC changes are smaller than the thresholded F1 changes. On Quo Vadis, removing semantic elements increases AUC by 0.0003, whereas removing the graph view and prior decreases it by 0.0006 and 0.0011. On Zenodo, removing semantic elements and the graph view decreases AUC by 0.0022 and 0.0035, while removing the prior increases it by 0.0006. The graph view therefore has the most consistent direction across datasets, although all effects remain small.

a) Single-seed leakage-safe component check.: As a strict-split calibration check, we also compared the full model with the variant that removes the preference objective on the zero-overlap Quo Vadis leakage-safe seed-7 split. The full model obtains Accuracy 0.8884, Precision 0.8421, Recall 0.6929, F1 0.7603, FPR 0.0445, and FNR 0.3071. Removing the preference loss obtains Accuracy 0.8794, Precision 0.8190, Recall 0.6772, F1 0.7414, FPR 0.0513, and FNR 0.3228. Thus, under this zero-overlap split, the preference objective improves thresholded detection behavior by increasing F1 by 0.0189 and Recall by 0.0157, while reducing FPR from 0.0513 to 0.0445 and FNR from 0.3228 to 0.3071. This check complements the balanced matched-ablation plan by showing that the preference objective helps reduce both false alarms and missed detections under the stricter Quo Vadis split; AUC remains comparable across the two variants.

Answer to RQ3. The matched ablation shows dataset-dependent component effects. The graph view contributes on both balanced splits, semantic elements show a clearer

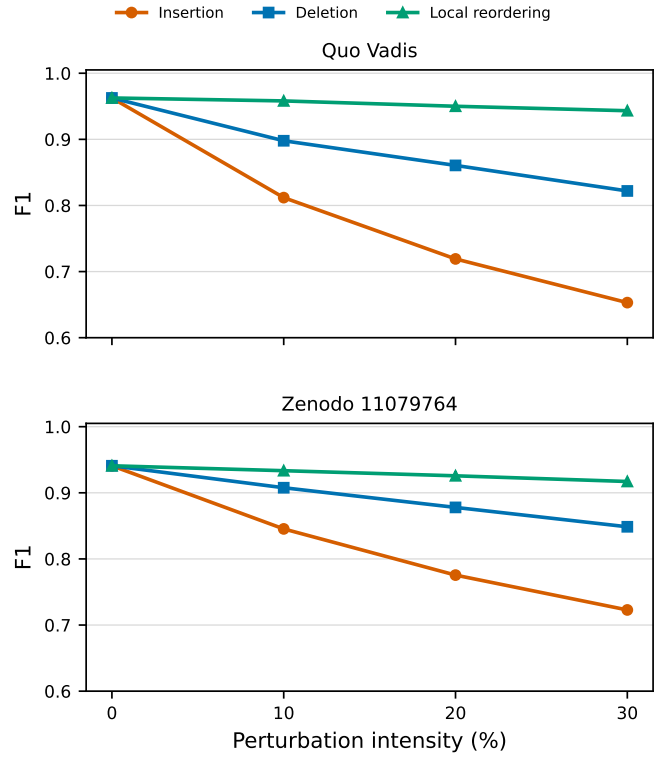


Fig. 2. Clean-trained seed-7 F1 trends. The 0% points are clean Full baselines; perturbed points reuse the clean checkpoint and threshold.

contribution on Zenodo but not Quo Vadis, and the interleaving prior has a small positive contribution on Quo Vadis but no positive marginal F1 contribution on Zenodo.

E. RQ4: Robustness

RQ4 evaluates perturbations applied to the balanced test splits while the seed-7 training data, validation-selected checkpoints, and decision thresholds remain clean and unchanged. Insertion adds API/syscall names sampled from the training-set benign-call distribution, deletion removes calls uniformly, and local reordering shuffles calls within bounded five-call windows. We evaluate each perturbation at 10%, 20%, and 30% intensity on both datasets.

Perturbations are deterministic per sample and condition. Insertion adds the rounded intensity fraction at seeded positions using the empirical benign training-call distribution; deletion samples positions without replacement while retaining at least one call. Local reordering selects the corresponding fraction of five-call windows and permutes calls only inside each selected window. Inputs that exceed the trained call capacity after insertion use the same deterministic uniform-coverage policy as model preparation. Each condition stores the transformed 7,500-record test set, configuration, metrics, and predictions.

Insertion causes the largest degradation on both datasets. At 30% intensity, F1 falls by 0.3096 on Quo

Vadis and 0.2181 on Zenodo, while AUC falls to 0.8429 and 0.7635, respectively. Deletion produces an intermediate effect: at 30%, AUC remains 0.9397 on Quo Vadis and 0.9071 on Zenodo despite F1 reductions of 0.1407 and 0.0923. Local reordering has the smallest effect; at 30%, F1 remains 0.9433/0.9171 and AUC remains 0.9890/0.9636 on Quo Vadis/Zenodo. Thus, the clean-trained model is comparatively stable to bounded local order changes but more sensitive to benign-call insertion. The 30% endpoint also clarifies the thresholded failure mode. On Quo Vadis, insertion retains Precision 0.9309 but lowers Recall to 0.5029, whereas deletion gives 0.9487/0.7251 and local reordering 0.9725/0.9157. On Zenodo, 30% insertion lowers Precision/Recall to 0.7074/0.7389, compared with 0.8229/0.8760 for deletion and 0.9037/0.9309 for local reordering. Insertion therefore produces missed detections on both datasets and a broader loss of separation on Zenodo.

Answer to RQ4. EsCapturer-full retains high AUC under local reordering and, to a lesser extent, deletion, whereas insertion increasingly degrades both thresholded detection and ranking as its intensity grows.

F. RQ5: Efficiency and Deployment Overhead

RQ5 uses the timing and memory counters stored by the two balanced Full runs. Both runs use CPU inference over the complete 7,500-record test fold. The frozen template extractor and graph construction are local preprocessing steps; no online LLM is called during training or inference.

Measurement	Quo Vadis	Zenodo
Training time	9.35 h	10.03 h
Semantic extraction	42.02 s	28.91 s
Behavior-unit construction	14.35 s	11.22 s
Graph construction	64.94 s	56.04 s
Test latency	13.41 ms/record	15.04 ms/record
Test throughput	74.6 records/s	66.5 records/s
Parameters	300,405	128,981
Peak RSS	1,718.7 MiB	1,687.0 MiB
Checkpoint size	1.30 MiB	0.54 MiB

These single-machine measurements are not a general deployment benchmark. Inference remains below 16 ms per record in both runs, whereas CPU training takes roughly 9–10 hours. Deployment can reuse the frozen 600.7 KiB Quo or 13.9 KiB Zenodo template cache.

Answer to RQ5. The measured CPU path processes 66–75 records/s without online LLM calls; training is substantially more expensive but remains an offline operation.

G. RQ6: Interpretability

RQ6 examines the graph-view gate stored by the two balanced Full artifacts. Interleaved behavior-unit pairs have $C_{ij} = 1$; non-interleaved pairs have $C_{ij} = 0$. We aggregate all test-fold pair weights after fixing the validation-selected checkpoint.

Dataset	Pair group	Pairs	Mean	Median
Quo Vadis	$C_{ij} = 0$	878,022	$< 10^{-9}$	$< 10^{-9}$
Quo Vadis	$C_{ij} = 1$	747,408	> 0.9999	1.0000
Zenodo	$C_{ij} = 0$	383,856	$< 10^{-9}$	$< 10^{-9}$
Zenodo	$C_{ij} = 1$	1,536,144	> 0.9999	1.0000

The gate is nearly binary on both datasets: graph weight is effectively zero for $C_{ij} = 0$ and one for $C_{ij} = 1$. This agrees with the encoded prior but also reveals saturation, so the weights provide limited nuance beyond the prior itself. They are not causal evidence of improved accuracy; RQ3 likewise finds dataset-dependent effects when the prior is removed.

Answer to RQ6. The gate reproduces the intended sequence/graph division, but its saturation limits instance-level interpretability and motivates calibration.

H. Threats to Validity

Construct validity. The balanced compact samples do not preserve source prevalence, and API names omit arguments, return values, and runtime objects needed to evaluate some semantic fields. Dataset-provided categories and families are used as labels, not independently validated malware lineages. The RQ4 perturbations model trace padding, omission, and bounded order noise synthetically; they do not cover adaptive attackers, argument mutation, or object obfuscation.

Internal validity. Exact duplicates, related windows, and shared source identities can inflate scores under ordinary random splits. The leakage-safe audit addresses this risk through conflicting-label exact sequence removal, exact deduplication, report/SHA group-disjoint splitting, and zero-overlap checks. Adapted and reimplemented baselines may still differ from their original systems. We restrict strict-split performance claims to Quo Vadis because only that audit has a complete three-seed repaired-model artifact set.

Conclusion validity. The main balanced benchmark and matched ablations use one seed-7 split, so we avoid significance and global-superiority claims. RQ2 adds three training seeds only for the Quo Vadis zero-overlap audit; robustness and gate-weight results remain single-seed analyses.

External validity. Both datasets contain Windows API traces from specific analysis environments. Results may not transfer to other sandbox versions, live endpoints, Linux system calls, packed malware, future families, or different deployment environments. Efficiency values come from one Apple CPU environment and should not be treated as hardware-independent latency bounds.

I. Summary of Findings

RQ1 shows competitive balanced-split detection without uniformly exceeding the local adapted baselines. RQ2 retains AUC 0.9168 ± 0.0121 across three Quo Vadis training seeds after exact- and group-disjoint reconstruction. RQ3 finds dataset-dependent component effects, while RQ4 shows that insertion is more damaging than deletion or local reordering. RQ5 records 66–75 CPU inferences per second, and RQ6 finds that the graph-view gate follows the interleaving prior but is nearly saturated.